



AI-MEMORY HACKATHON · SAN FRANCISCO

# Build the LLM Wiki Your Agents Actually Remember

*"Instead of just retrieving from raw documents at query time, the LLM incrementally builds and maintains a persistent wiki" — Karpathy*



# Redis 101

The world's fastest in-memory data platform, built for AI.

**<1ms**

typical read/write  
latency

**62%**

faster than the  
#2 vector DB

**10+**

data types  
(strings, hashes,  
vectors & more)

**1**

pip install  
away

What Redis does for your AI stack:

Vector Search

Semantic Caching

Agent Memory

Session State

Semantic Router

Full-Text Search

Redis is NOT just a cache. It's a full-featured database: run vector similarity search, full-text search, pub/sub, streams, geospatial, time series — all in one.

# Search in Redis

powered by RedisVL — the Python vector library for Redis

## `pip install redisvl`

- Dedicated Python client for Redis as a vector DB
- Built-in support for HNSW & FLAT index algorithms
- Automatic embedding generation
- Semantic caching to cut LLM costs
- Reranking, hybrid search, and routing out-of-the-box
- Fully async — non-blocking in agent pipelines

## Vector Search

KNN & range queries  
over dense embeddings

## Hybrid Search

Combine vector + full-text  
(BM25) in one query

## Semantic Cache

Skip re-calling the LLM  
for similar prompts

## Semantic Router

Route user intent to  
correct tool/agent

# Vector Search in 10 Lines

RedisVL — index creation, embedding + search

```
from redisvl.index import SearchIndex
from redisvl.query import VectorQuery
from redisvl.utils.vectorize import OpenAITextVectorizer

# 1. Define your schema
schema = {
    "index": {"name": "docs", "prefix": "doc"},
    "fields": [
        {"name": "content", "type": "text"},
        {"name": "embedding", "type": "vector",
         "attrs": {"dims": 1536, "algorithm": "hnsw",
                  "distance_metric": "cosine"}}
    ]
}

# 2. Create index & load data
index = SearchIndex.from_dict(schema)
index.connect("redis://localhost:6379")
index.create(overwrite=True)

# 3. Vectorize & search
vectorizer = OpenAITextVectorizer()
query_vec = vectorizer.embed("What is Redis?")

results = index.query(
    VectorQuery(query_vec, "embedding", num_results=5)
)
```

## What you get

### HNSW indexing

Sub-millisecond ANN at  
hundreds of millions of vectors

### Hybrid queries

Mix vector KNN with  
full-text or numeric filters

### Async native

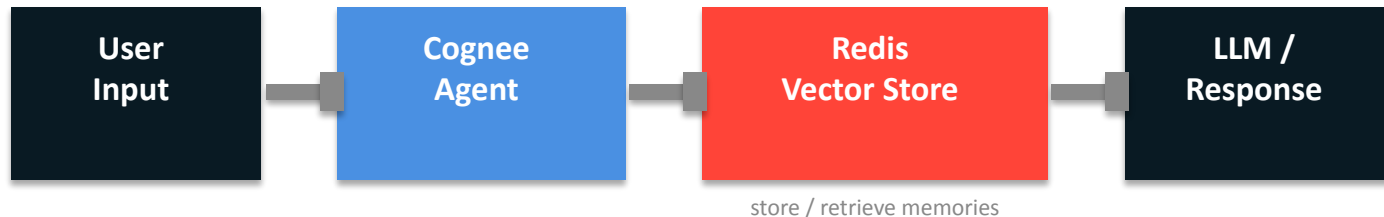
Non-blocking in  
FastAPI / async agents

### Auto-sync

Index updates on  
write — no reindex needed

# Cognee + Redis

Plug Redis in as Cognee's vector memory backend — one import away



```
import cognee
from cognee.infrastructure.databases.vector \
    import get_vectoradb_config

# Point Cognee at Redis as vector backend
config = get_vectoradb_config()
config.vector_db_provider = "redis"
config.vector_db_url      = "redis://localhost:6379"

# Now use Cognee normally — Redis handles the memory
await cognee.add(documents)
await cognee.cognify() # builds the knowledge graph
results = await cognee.search("What did we discuss?")
```

## Why Redis for Cognee memory?

- Persistent memory across agent runs
- Sub-ms similarity search at scale
- Semantic deduplication built-in
- Works with any embedding model
- Free cloud tier — no infra setup



**IRIS**

by Redis

# Something big is coming.

Redis Iris — an intelligent layer that brings native AI reasoning to your Redis data. Launching this Monday.

Stay tuned — Monday launch

# Scale your startup faster

This program offers everything growing businesses need to build fast apps fast.

To qualify, you must be a venture-backed, seed or below startup.



## *Redis Startup Program*

For more information:  
[redis.io/startups](https://redis.io/startups)



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## Free Redis for qualifying startups

Angel, Pre-seed, and Seed stage startups can take advantage of exclusive benefits to help your business build fast apps and scale earlier.

- \$25k Redis Cloud credit for 6 months—usage limits apply
- Early access to new Redis features and products



## Access to Redis support

Partner with our experts, mentors, and fellow startups to design apps that succeed at scale, perform seamlessly, and more.

- Dedicated customer support, including monthly office hours and a basic support plan
- Learn from our industry-leading applied AI team



## Build your brand and go to market

Use our network of customers, partners, and industry connections to grow your go-to-market strategy.

- Access co-marketing opportunities and lead generation programs
- Meet potential investors and customers

# Win the Best Use of Redis Prize

Use Redis Cloud, RedisVL, or the Cognition Redis Adapter in your hack

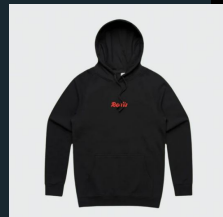
## AirPods Pro

(one per team member)



## Redis Hoodies

(one per team member)



## 10K Redis Cloud

### Credits

(for the team)



Redis Cloud

### To qualify:

- Use Redis Cloud, RedisVL, or the Cognition Redis Vector Adapter (demonstrated in demo + GitHub)
- Build something new today — no plagiarism
- Strong preference for interesting Redis use beyond basic caching





## AI Resources

● Redis VL



● Redis Cognition Adapter



● Redis Agent Skill Repo

